

JKBOSE 2026

CLASS 11TH

HYDROCARBONS

LECTURE-09 CHEMISTRY

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JKBose Wallah

Topics To Covered



- 1 Structure of Benzene**
- 2 Aromaticity**
- 3 Preparation of Benzene**
- 4 Physical Properties**
- 5 Chemical Properties**
- 6 Directive Influence of a Functional**

WELCOME MY DEAR

STUDENTS.

Vishal Sir Loves you so much.

STRUCTURE OF BENZENE.

AROMATICITY.

a) Cyclic

b) Conjugation

c) $(4n+2)\pi e$

AROMATIC

a) Cyclic

b) Conjugation

c) $4n \pi e$

ANTIAROMATIC



ORDER OF STABILITY.

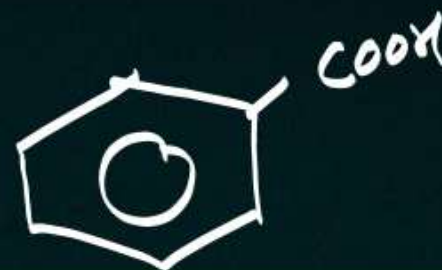
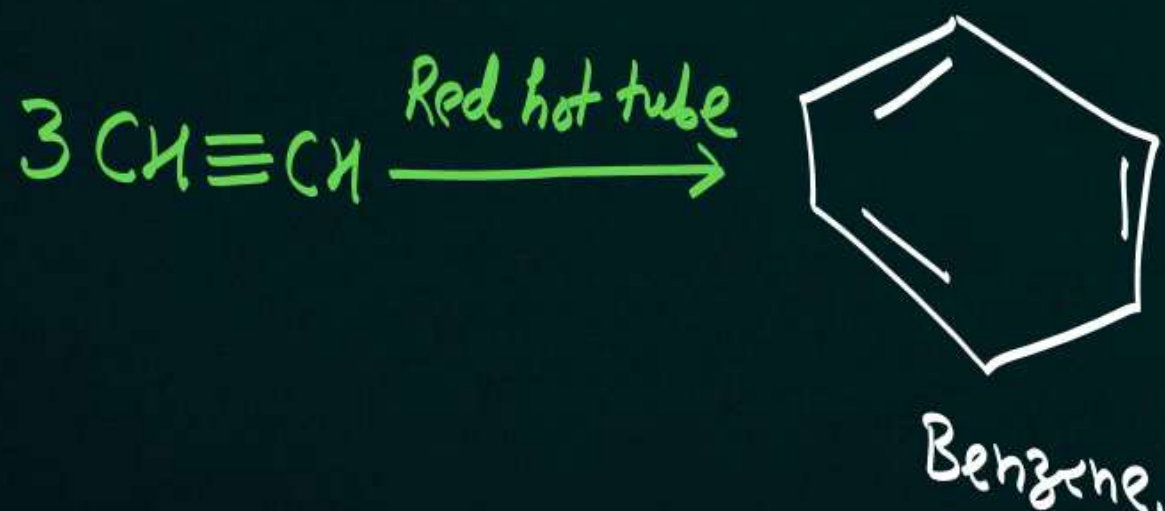
AROMATIC > Non
AROMATIC > Anti
aromatic.



Preparation of Benzene

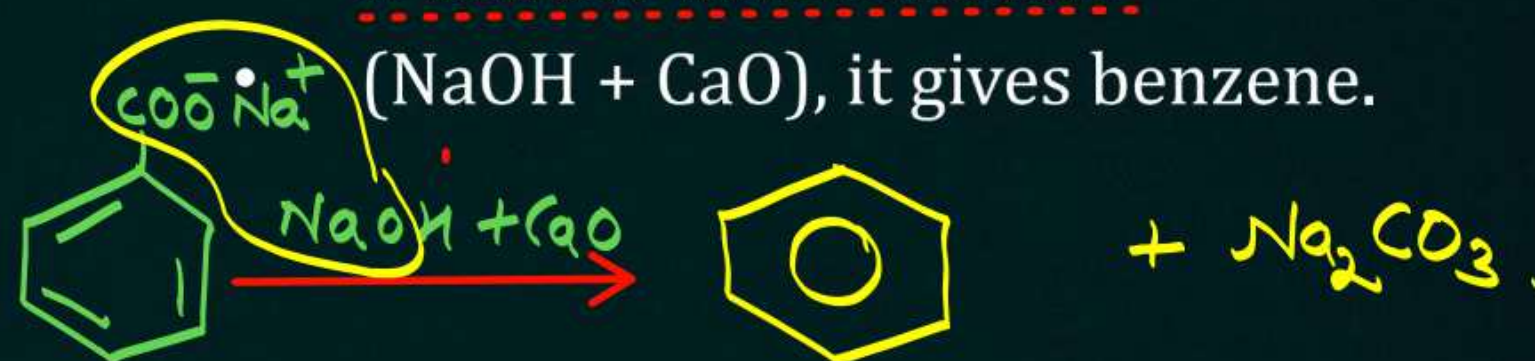
(a) Cyclic Polymerisation of Ethyne

- When acetylene is passed through a red hot metallic tube, cyclic polymerisation takes place and benzene is formed.



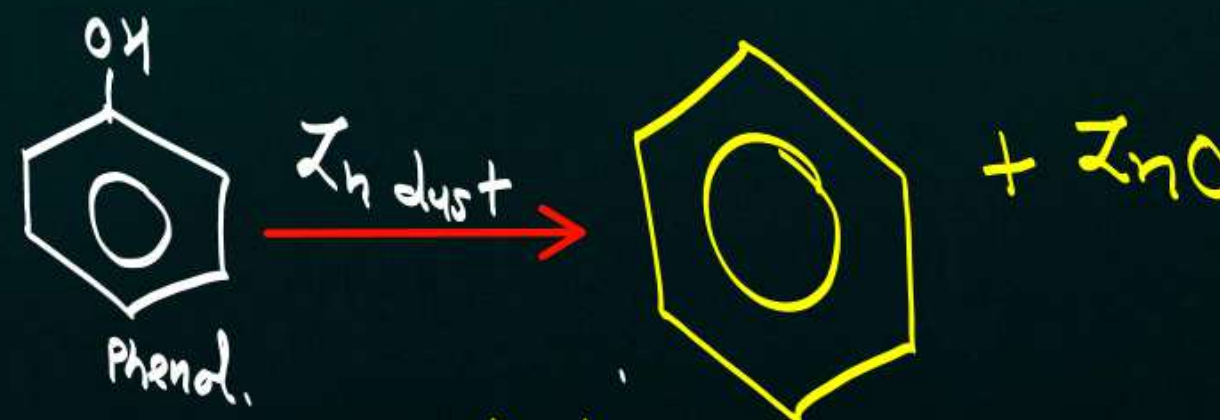
(b) Decarboxylation of Aromatic acids

- When sodium salt of benzoic acid is heated with sodalime



(c) Reduction of Phenol

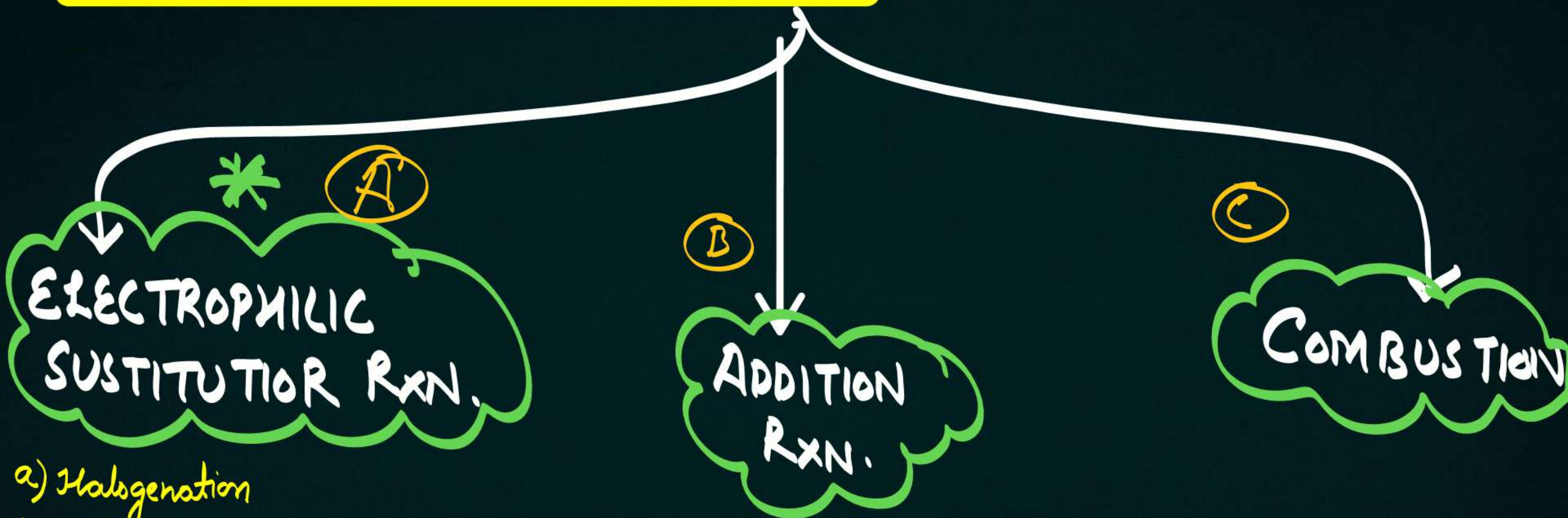
Phenol on reaction with Zn dust produces benzene.



Physical Properties of Benzene

- (i) Aromatic hydrocarbons being non-polar molecules are insoluble in water but are readily miscible with organic solvents.
- (ii) They are colourless liquids or solids that have characteristic aroma, e.g. naphthalene balls.
- (iii) They burn with sooty flame.

Chemical Properties of Benzene

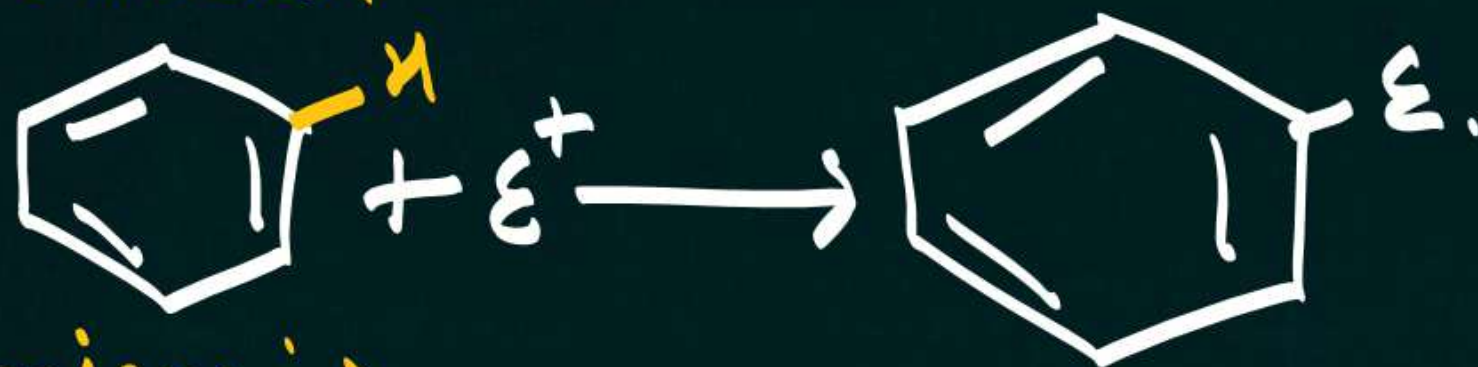


- a) Halogenation
- b) Nitration
- c) Sulphonation
- d) Alkylation
- e) Acylation

Chemical Properties of Benzene

(A) Electrophilic Substitution Reactions

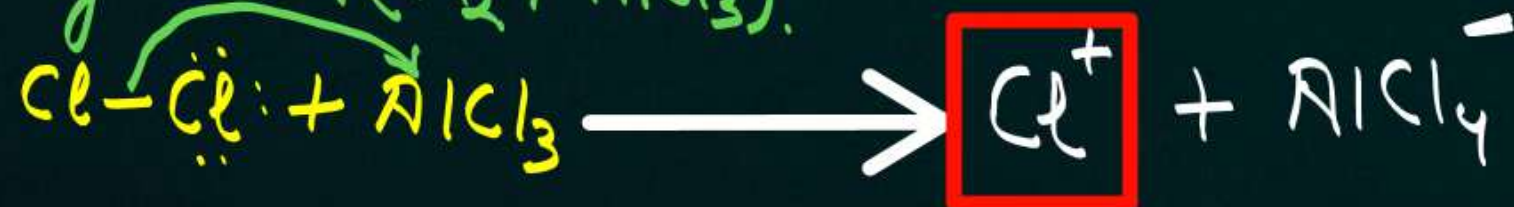
General reaction



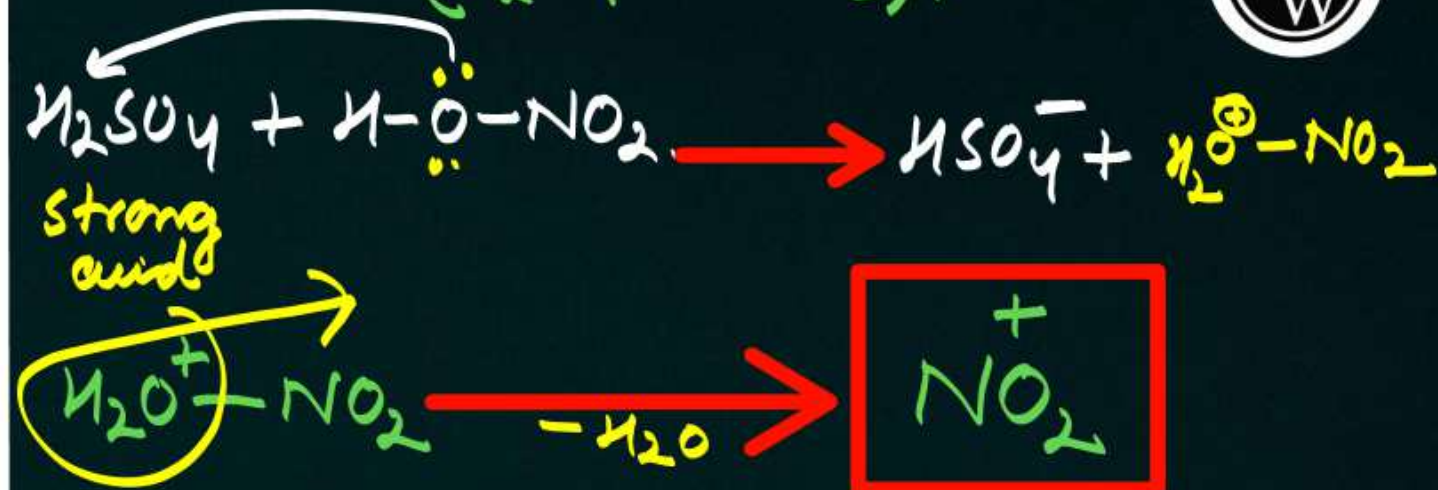
Mechanism:

1) Generation of Electrophile.

a) Halogenation ($X_2 + AlCl_3$).



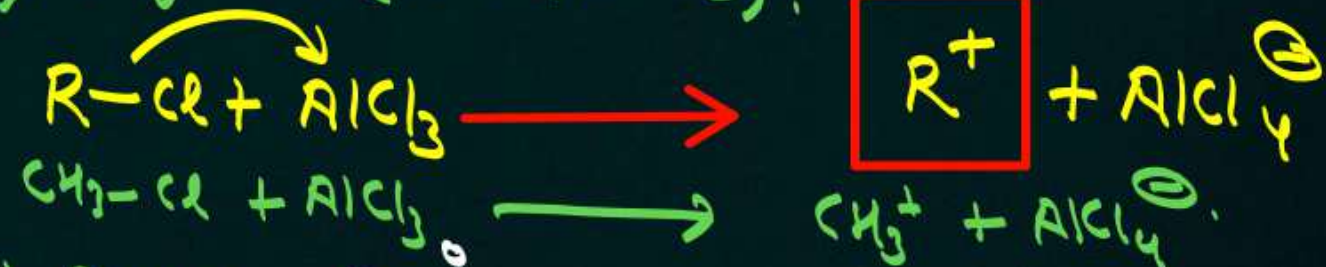
b) Nitration ($H_2SO_4 + HNO_3$).



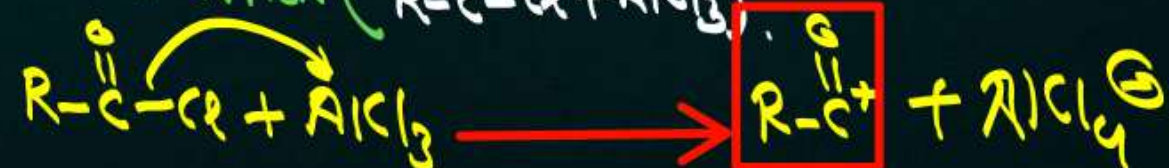
c) Sulphonation (H_2SO_4 conc. or $H_2S_2O_7$ (oleum))



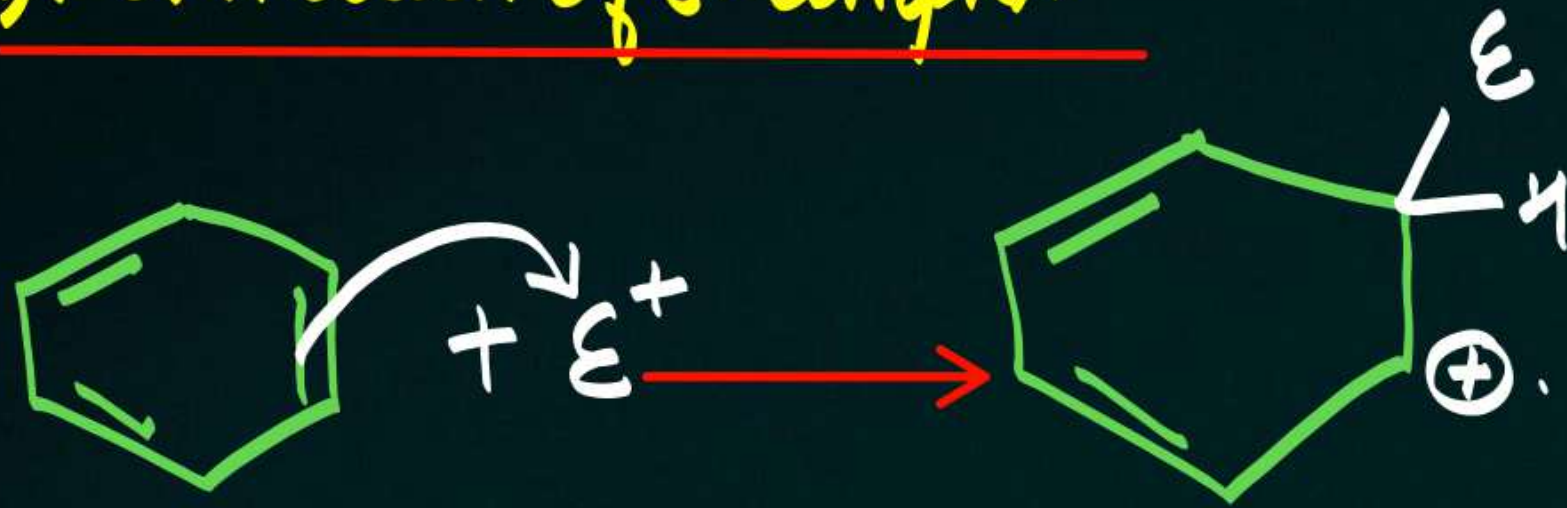
d) Alkylation ($R-X, AlCl_3$).



e) Acylation ($R-C(=O)-Cl + AlCl_3$).

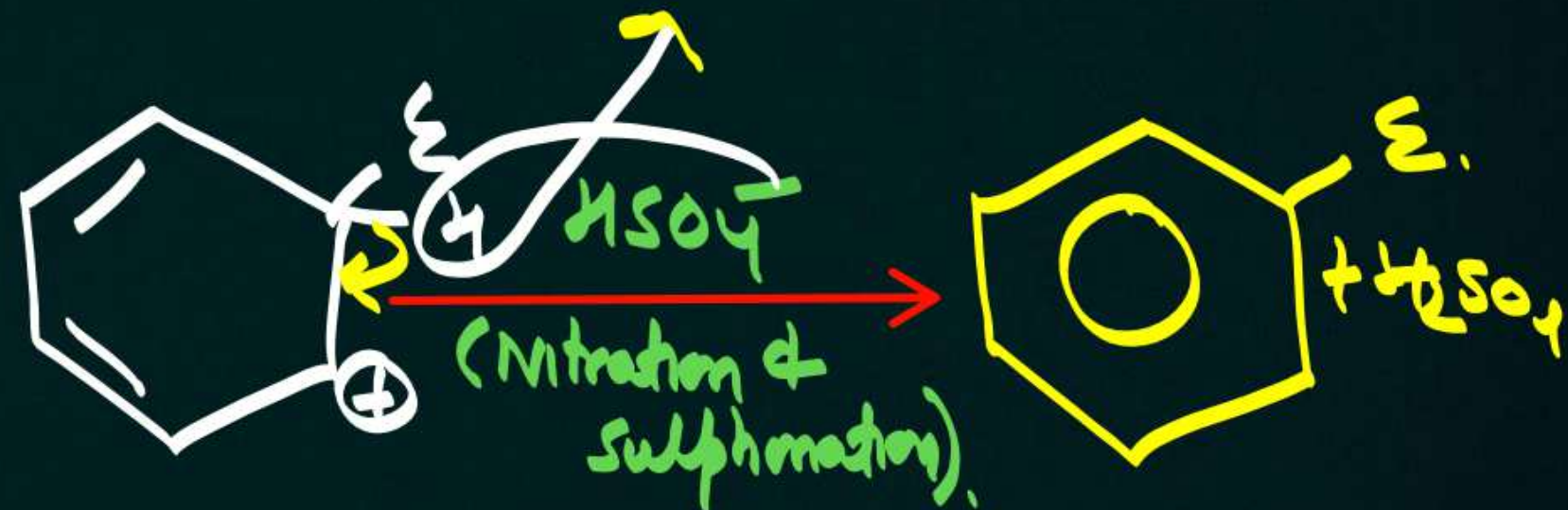
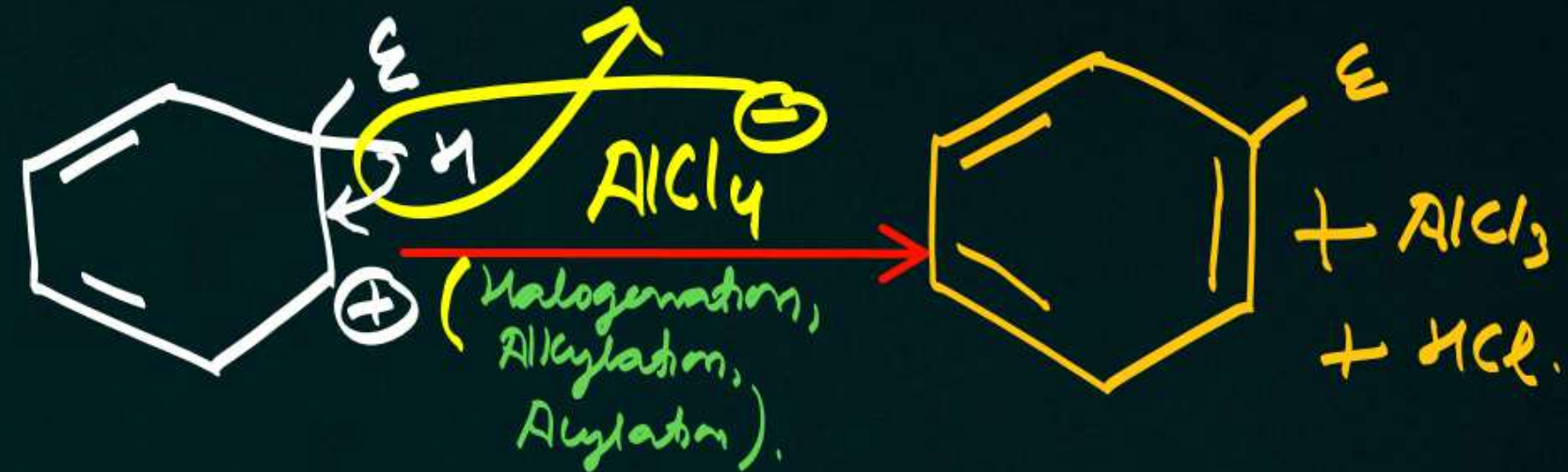


2) Formation of σ -complex.



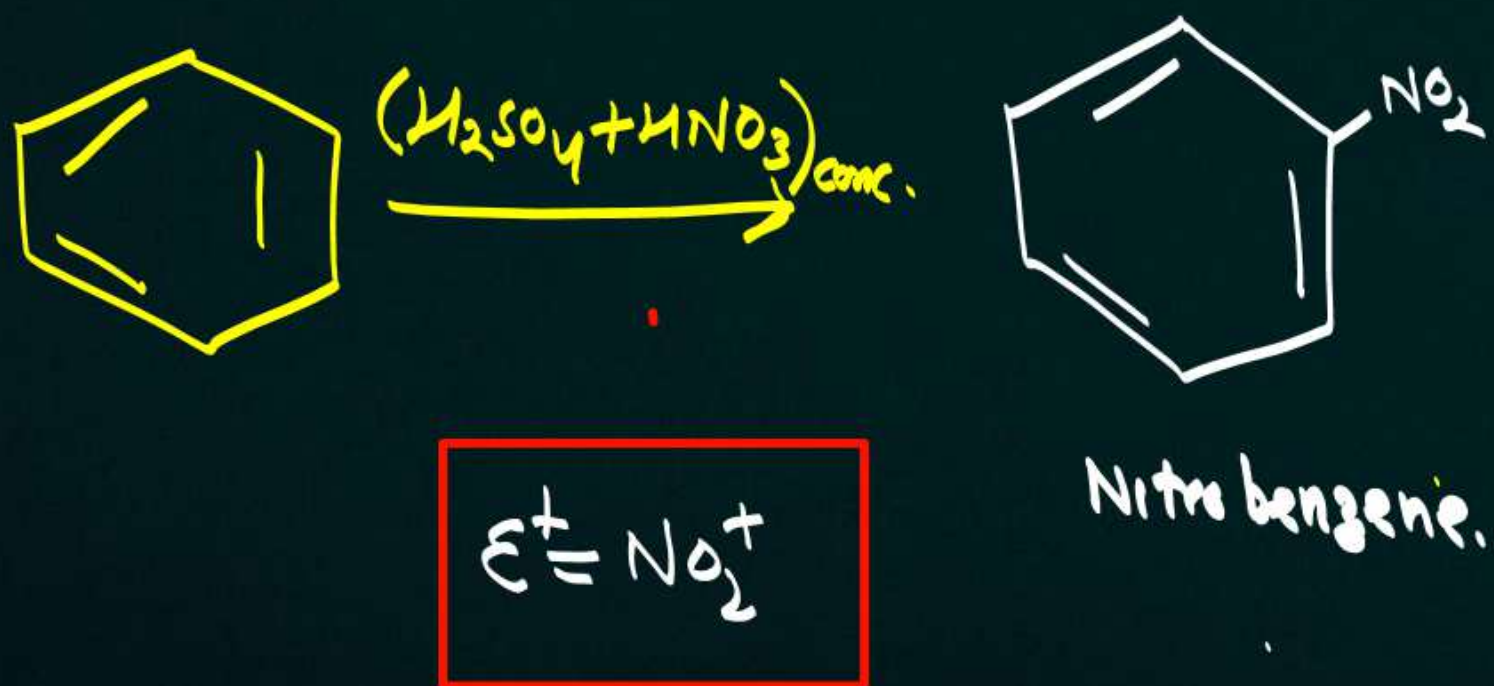
σ -Complex
or
Arenium ion
or
Wheland complex.

3) Abstraction of proton.

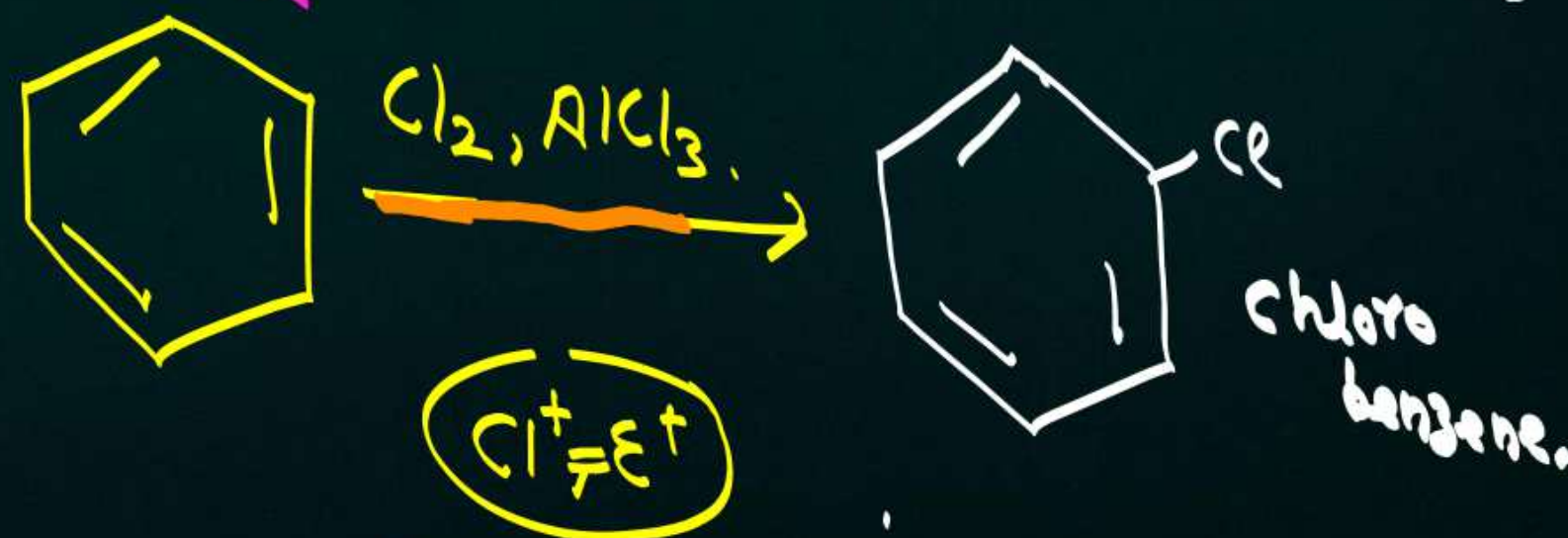


Chemical Properties

(i) **Nitration** Benzene undergoes nitration when treated with conc. HNO_3 in the presence of conc. sulphuric acid.



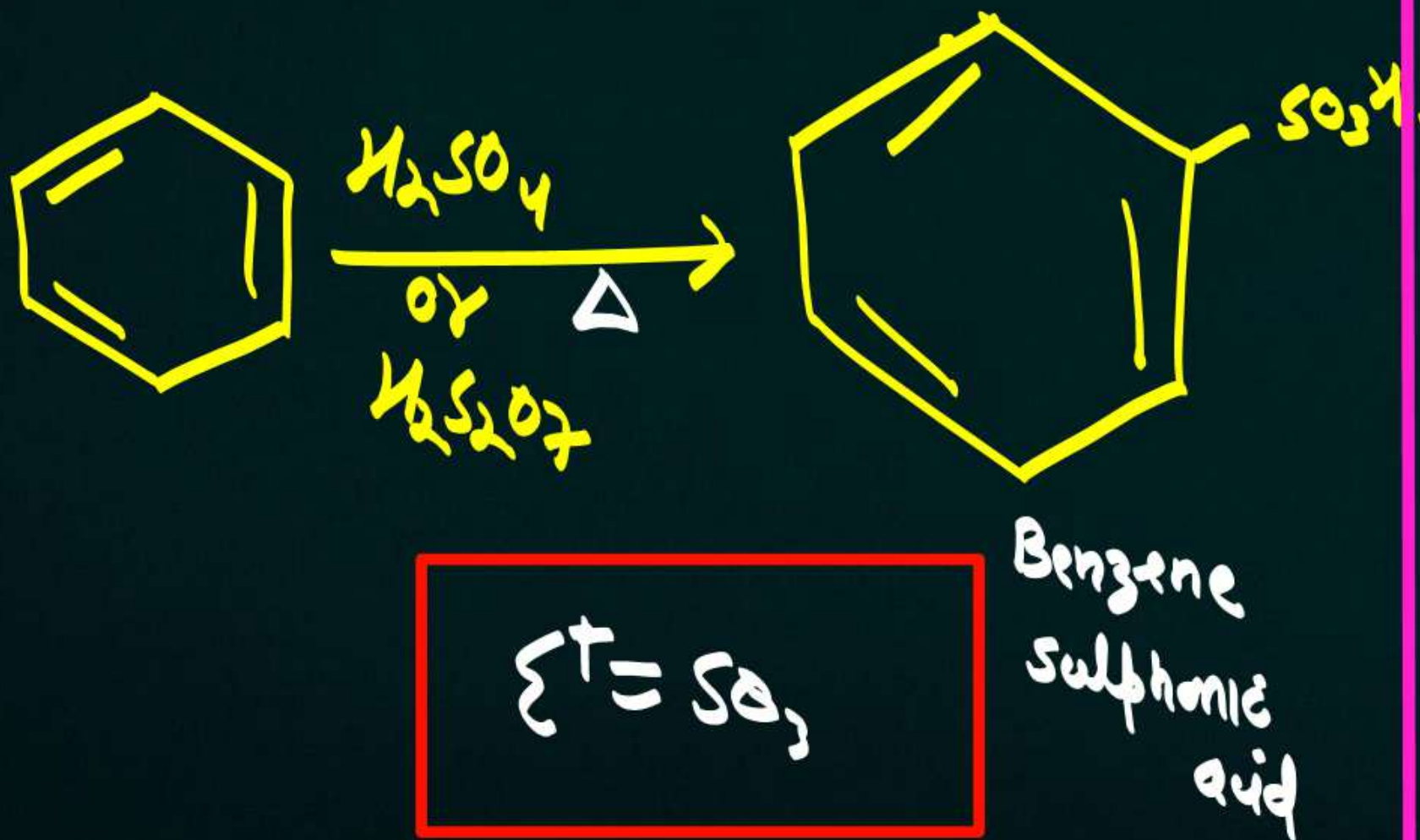
(ii) **Halogenation** Benzene undergoes chlorination or bromination when it is treated with chlorine or bromine in the presence of Lewis acid catalyst, aluminium chloride or ferric chloride or ferric bromide and in the absence of light. ($\text{AlCl}_3, \text{FeCl}_3, \text{FeBr}_3$)



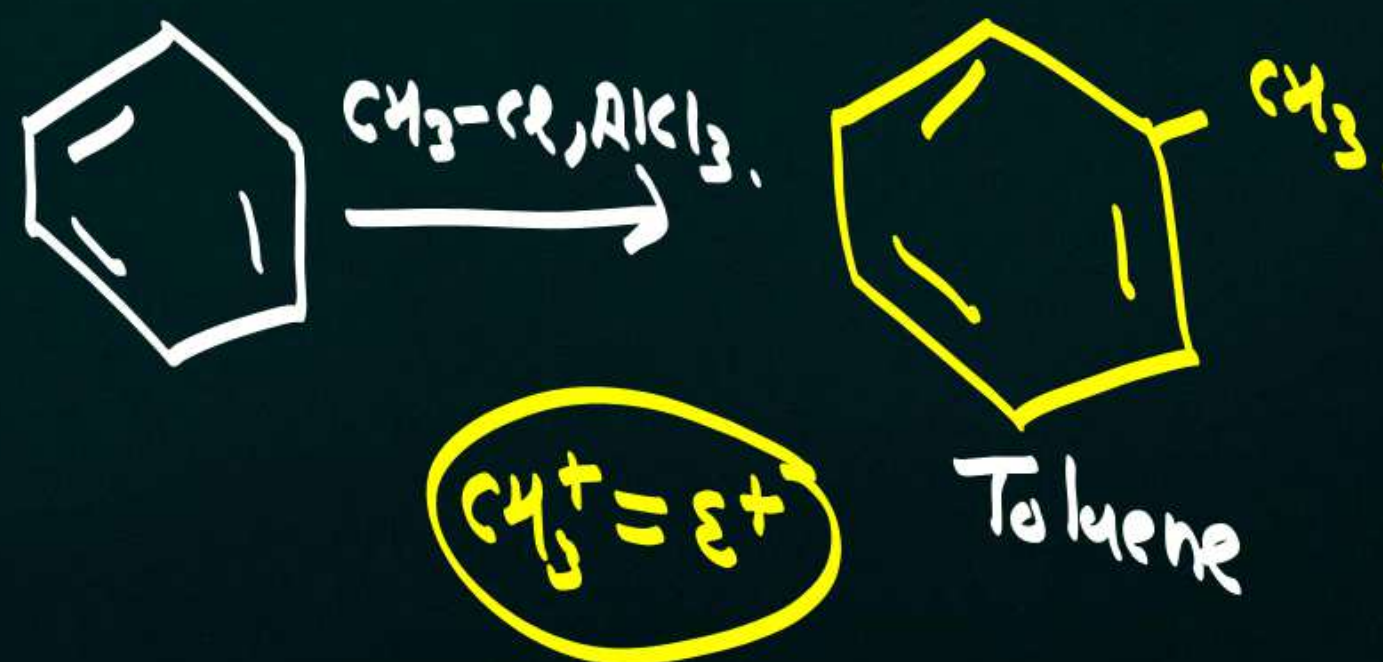
Chemical Properties



(iii) **Sulphonation** It is done by heating benzene with fuming sulphuric acid (oleum).



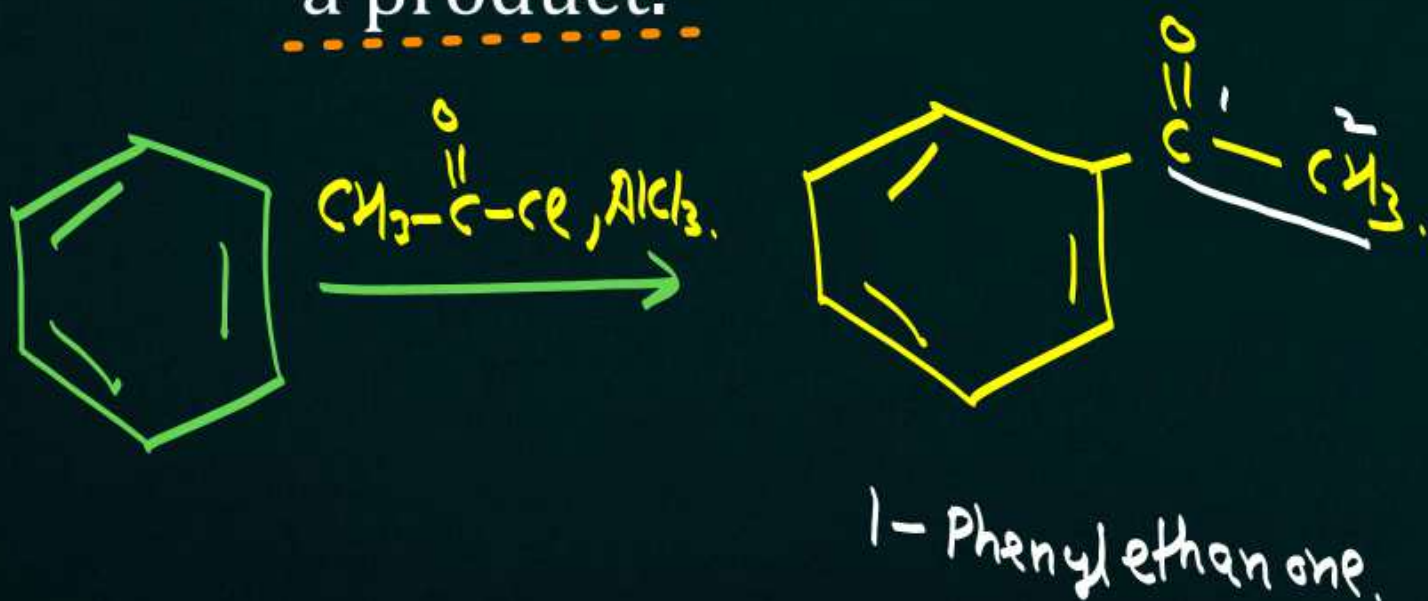
(iv) **Friedel-Crafts alkylation** When benzene is treated with alkyl halide in the presence of anhyd. AlCl_3 , it gives alkyl benzene. This reaction is called Friedel-Crafts alkylation.



Chemical Properties

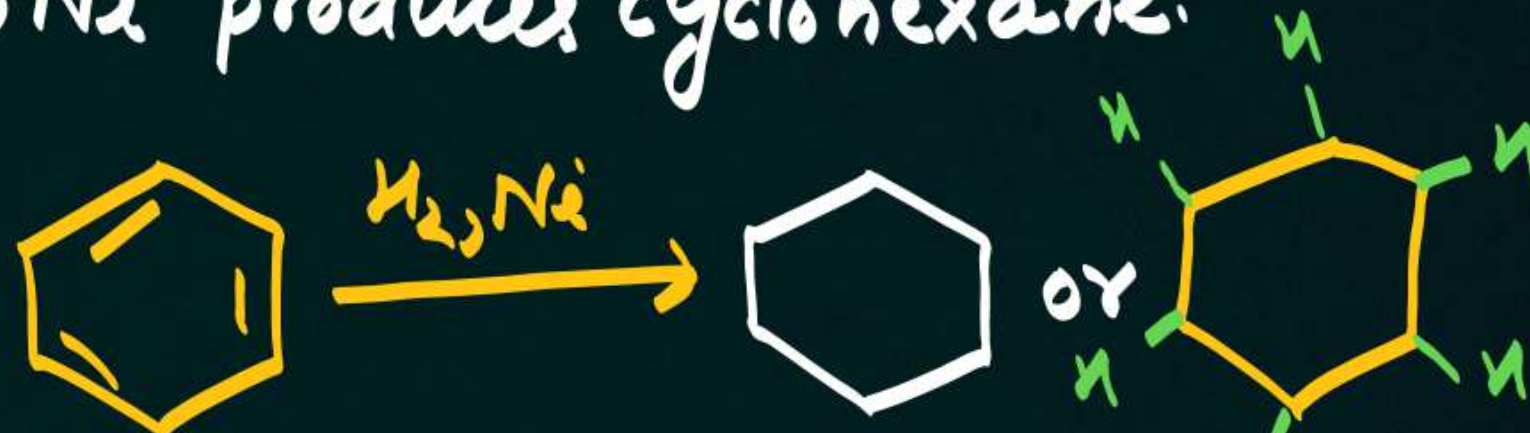


(v) **Friedel-Crafts acylation** When benzene is treated with ~~halide~~ ^{acyl} halide or acid anhydride in the presence of a Lewis acid (like $AlCl_3$), acyl benzene is obtained as a product.

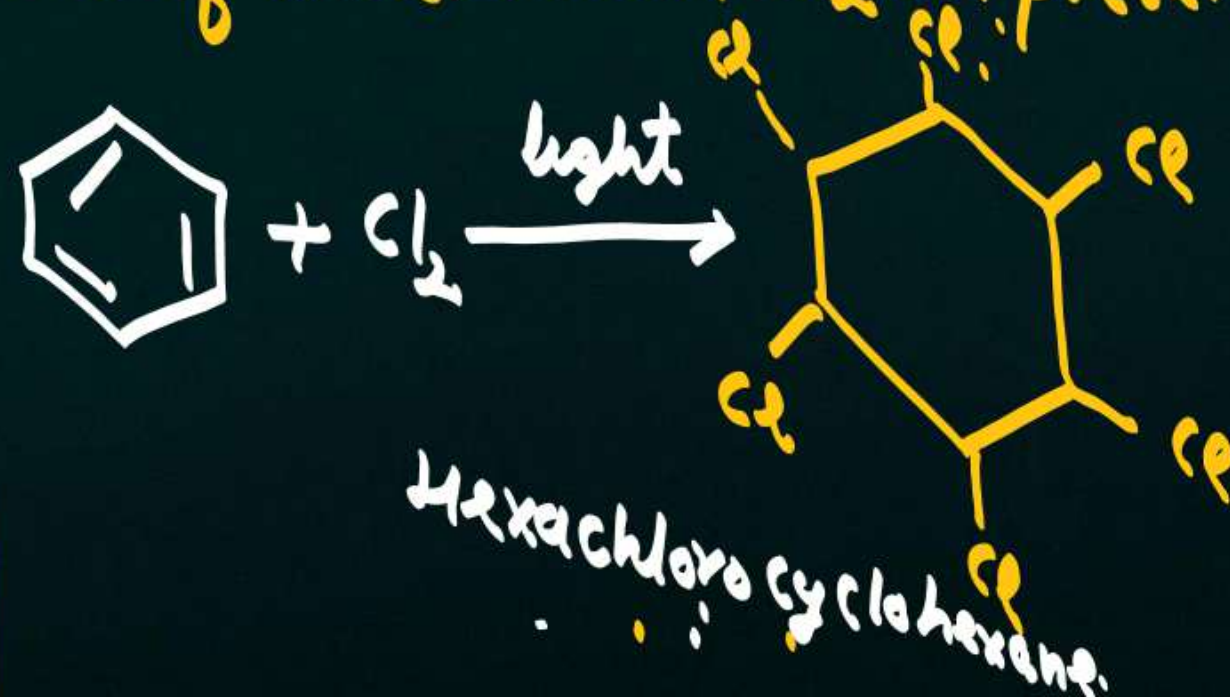


(B) Addition Reactions

Benzene on treatment with H_2 , Ni produces cyclohexane.



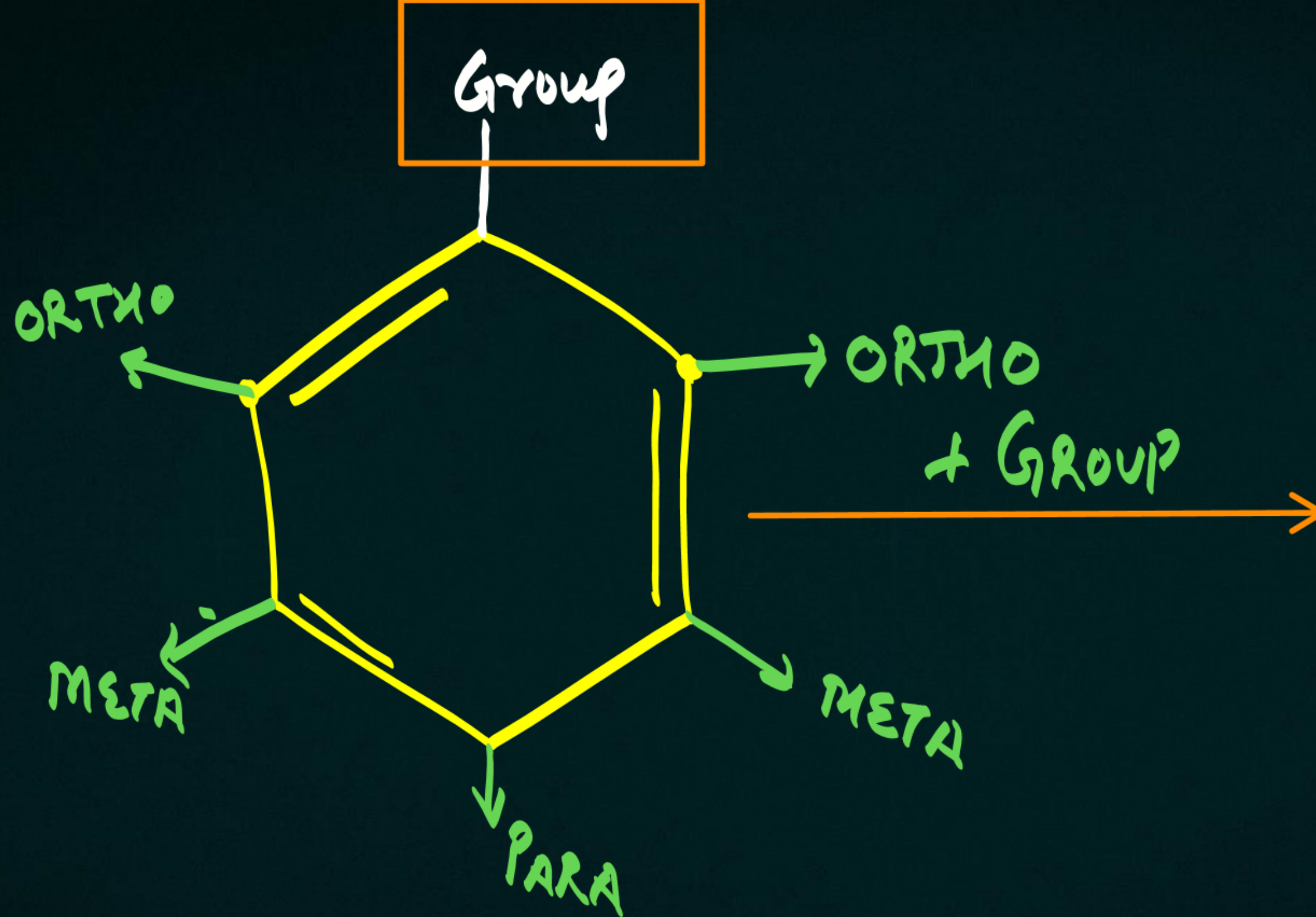
But if treated with Cl_2 in presence of light.



Chemical Properties

(C) **Combustion** $\rightarrow$ Benzene when heated in presence of O_2 produces $CO_2 + H_2O$.





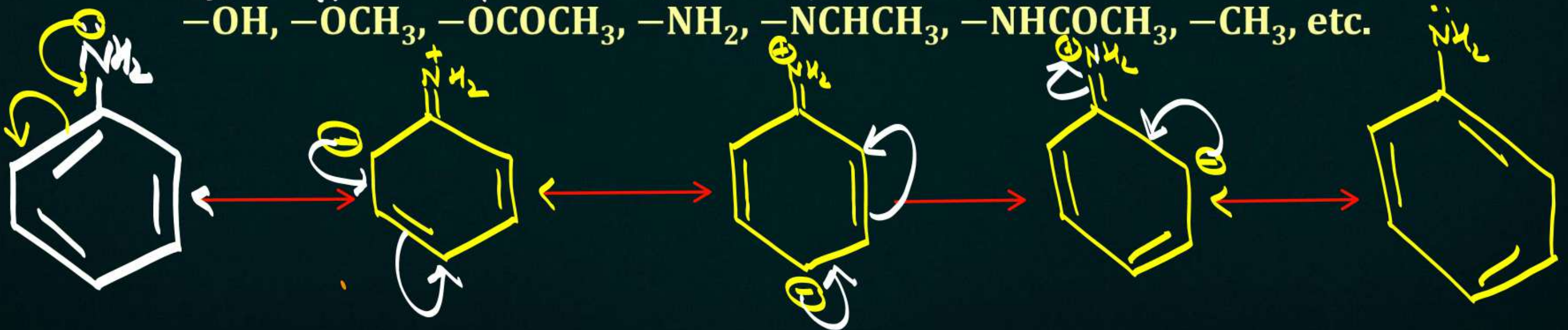
Directive Influence of Substituents on Benzene Ring

The ability of a group already present in the benzene ring to direct the incoming group to a particular position is called the directive influence of groups.

1. Ortho and para-directing activators. These groups release electrons and activate the benzene ring. These groups direct the incoming groups to ortho and para positions. The common examples are:

(+R)

$-\text{OH}$, $-\text{OCH}_3$, $-\text{OCOCH}_3$, $-\text{NH}_2$, $-\text{NCH}_2\text{CH}_3$, $-\text{NHCOCH}_3$, $-\text{CH}_3$, etc.

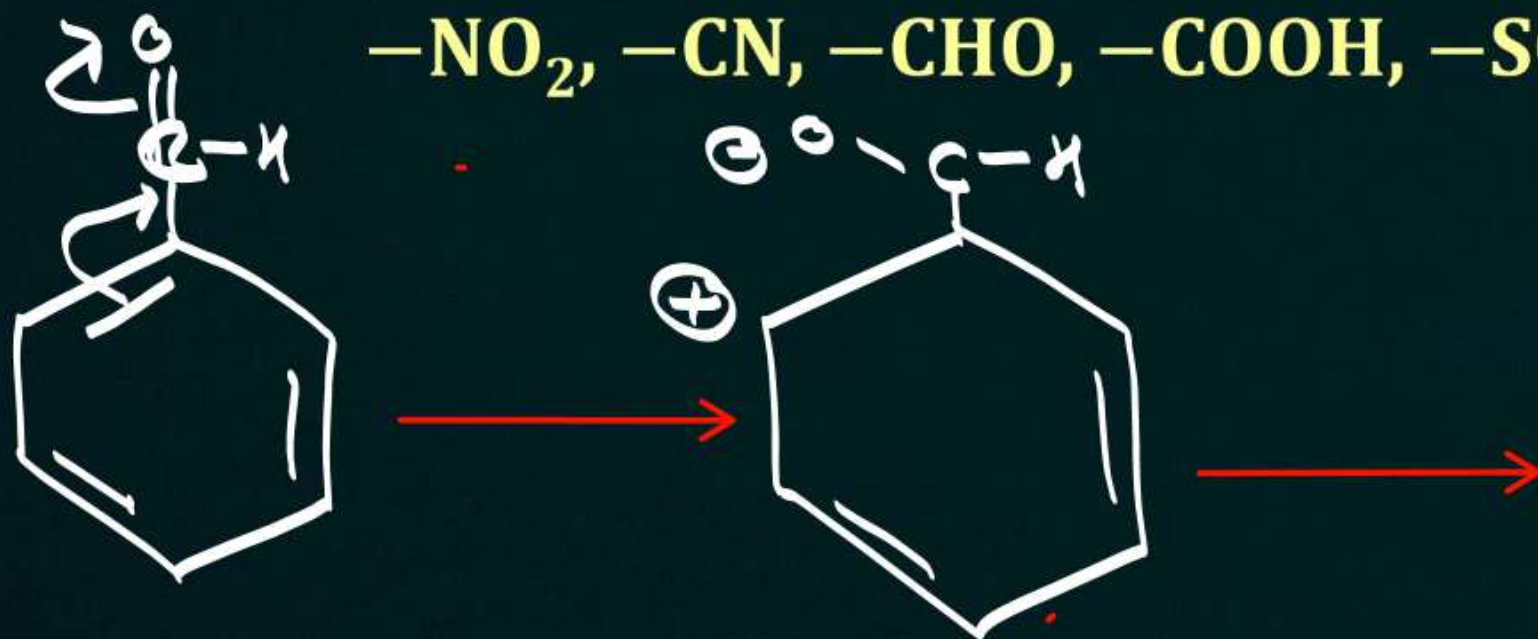


Directive Influence of Substituents on Benzene Ring

2. Meta directing deactivating groups. These groups withdraw electrons from the benzene ring and deactivate it. These groups direct the incoming group to meta positions. The common examples are:

(-R)

$-\text{NO}_2$, $-\text{CN}$, $-\text{CHO}$, $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{COCH}_3$.

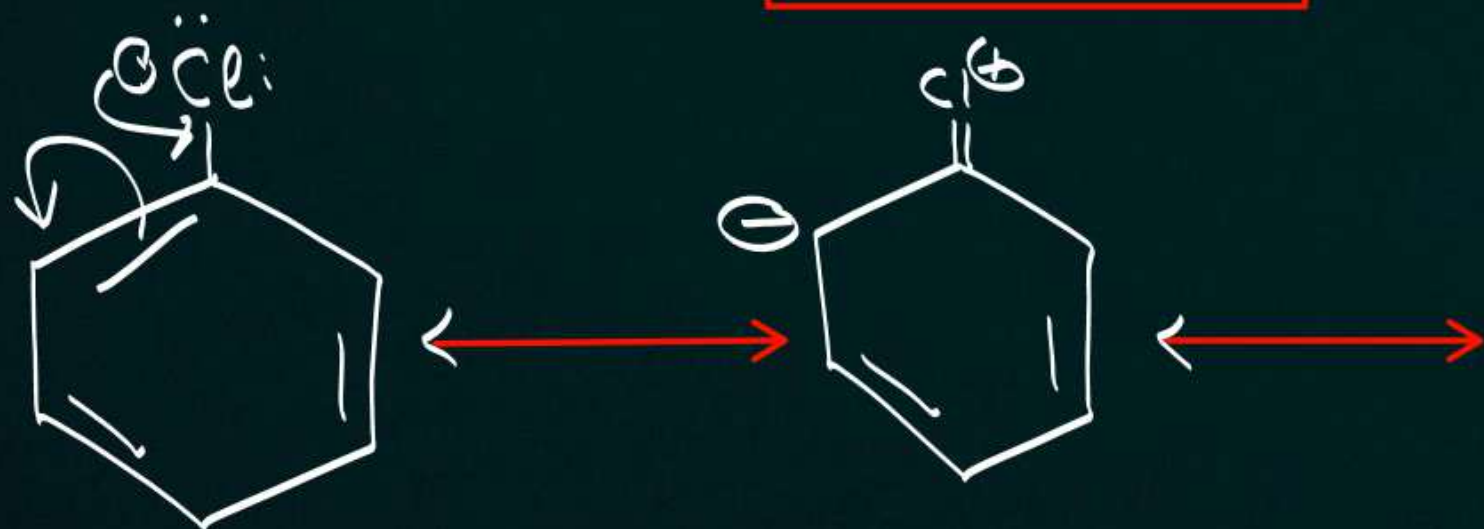


Directive Influence of Substituents on Benzene Ring

3. Ortho and para directing deactivating groups. These groups withdraw electrons from the benzene ring and deactivate it. However, these direct the incoming group to *ortho* and *para* positions. The common examples are:

Halogens: $-F, Cl, Br, I.$

$+R$



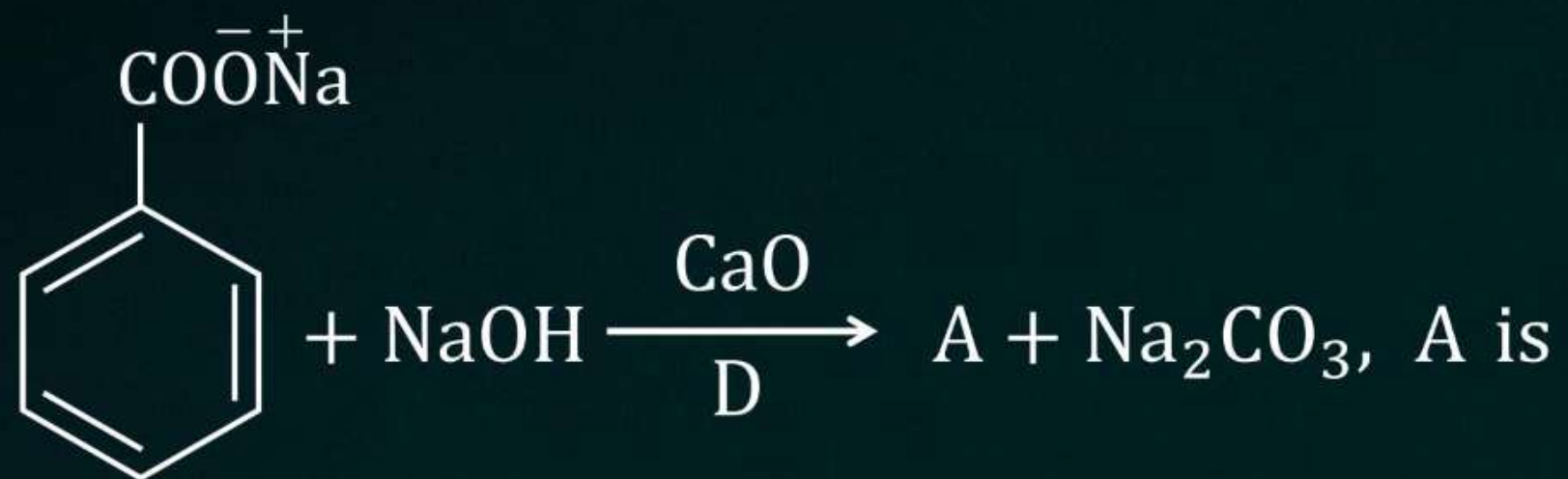
a) o-p directing activators.

b) m-directing deactivity.

c) o-p directing deactivator.

$-I \rightarrow$ withdraw e^-
 $+R \rightarrow$ Donate e^-

QUESTION



- 1 C₆H₆
- 2 C₆H₅ - Na
- 3 C₆H₅ - ONa
- 4 C₅H₅ - Na

QUESTION

Attack of electrophile on benzene results in the formation of

- 1 σ complex
- 2 Arenium ion
- 3 π complex
- 4 Both (1) and (2)

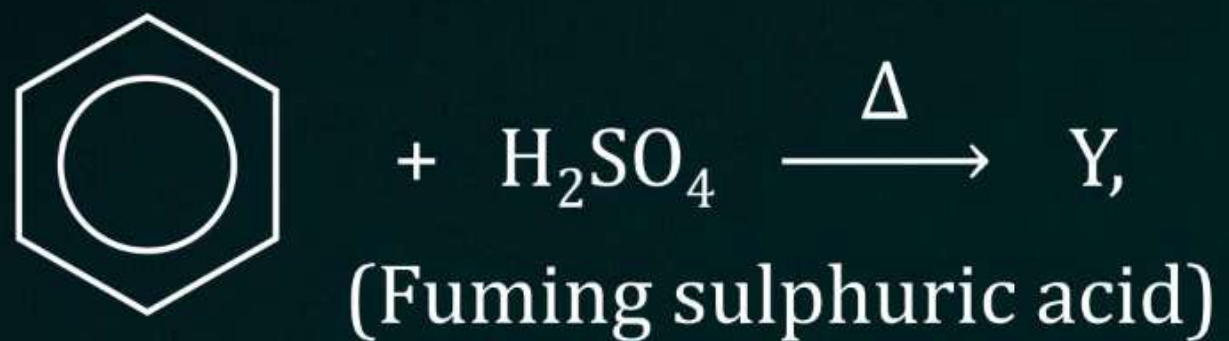
QUESTION

Which of the following set of functional groups is *meta-directing* group?

- 1 $-\text{NO}_2, -\text{NH}_2, -\text{COOH}, -\text{COOR}$
- 2 $-\text{NO}_2, -\text{CHO}, -\text{SO}_3\text{H}, -\text{COR}$
- 3 $-\text{CN}, -\text{CHO}, -\text{NHCOCH}_3, -\text{COOR}$
- 4 $-\text{CN}, -\text{NH}_2, \text{NHR}, -\text{OCH}_3$

QUESTION

In the reaction,



QUESTION

The aromatic compound among the following is



Thank You My.

#XUNV

PROMISE →

"HUM TUDAA NAHI HUE
MAIN

MILENGE RESULT CELEBRATION

Me 

Class 11th Syllabus.

COMPLETE



• Thank You ^{so much} for Memories.

Thank
You